

Home Robot Motor Imagery Interaction and Emotional Awareness based on Multimodal EEG Feature Extraction

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Abstract—With the rapid advancement of Brain-Computer Interface (BCI) technology, real-time control of external devices via electroencephalogram (EEG) signals has become feasible, offering new avenues for building more natural and efficient human-computer interaction systems. To address these issues, this paper proposes a Rule-based Motor Imagery Interaction Model (Rule-MII Model) that combines rule matching with multi-feature fusion. By leveraging discrete wavelet transforms, time-frequency domain analysis, and symbolic pattern mining, the model extracts multidimensional EEG features and constructs a standard rule set for recognizing users' motor intentions and emotional states. The proposed model achieves classification accuracies of 91.0% and 92.7% on the public BCI2a and DEAP datasets for motor imagery and emotion recognition tasks, respectively. Experimental results demonstrate that the proposed method exhibits strong accuracy and responsiveness in multimodal EEG signal modeling and mixed-reality-based human-robot interaction control, providing a promising solution for immersive and context-aware human-robot collaboration.

Index Terms—Motor Imagery, Human Robot Interaction, EEG Feature Extraction, Real Time, Data Mining.

I. INTRODUCTION.

With the continuous advancement of Mixed Reality (MR) and immersive interaction technologies, human-machine collaboration is evolving from traditional “explicit input-driven” paradigms toward more multimodal, embodied, and context-aware forms of interaction. In this transition, the Brain-Computer Interface serves as a critical bridge for achieving deep human-machine integration, enabling users to control systems implicitly through electroencephalogram signals. EEG directly reflects neural activity and offers advantages such as high temporal resolution, low hardware cost, portability, and non-invasiveness, making it especially suitable

for MR scenarios that require real-time responsiveness and natural control. Unlike speech or gesture inputs, EEG can capture users' cognitive and emotional fluctuations without any overt physical movement, thus creating new possibilities for natural and immersive interaction experiences.

Compared with other neuroimaging techniques such as functional magnetic resonance imaging (fMRI), EEG demonstrates greater practicality and portability in MR environments. Although fMRI provides higher spatial resolution, its large equipment size and high latency (typically on the order of seconds) make it unsuitable for real-time interaction. In contrast, EEG records brain signals on a millisecond timescale and can achieve stable acquisition in mobile settings through wearable devices. Moreover, EEG's low-latency characteristics give it notable advantages in emotion recognition, attention tracking, and intention prediction, all of which are essential for supporting dynamic adaptability and contextual awareness in MR systems.

Among various EEG paradigms, Motor Imagery (MI) and Emotion Recognition are particularly important for representing user intentions and affective states. MI involves the mental rehearsal of specific movements, activating brain regions similar to those used during actual motion execution [1], while emotional states provide personalized and context-sensitive cues for adaptive interaction systems. However, EEG signals are inherently nonlinear, non-stationary, and have a low signal-to-noise ratio [2], making robust recognition in MR scenarios challenging. To address these issues, this paper proposes a Rule-based Motor Imagery Interaction Model (Rule-MII) that integrates rule matching with multi-feature fusion. The proposed model enables precise recognition and mapping of users' motor intentions and emotional states. Experimental results on the BCI2a and DEAP datasets demonstrate superior

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performance, offering a feasible and efficient solution for building immersive, embodied EEG-driven interaction systems.

II. RELATED WORK.

Motor imagery classification and emotion recognition are critical research directions within the field of EEG-based brain-computer interfaces. Venkata Chunduri et al. proposed a multi-scale spatiotemporal attention network that leverages self-attention mechanisms to classify motor imagery EEG signals, demonstrating high classification accuracy [3]. In contrast, the No-Filter EEG model introduced by Emre Ari and Ertuğrul Taçgın processes raw EEG data directly, improving classification real-time performance [4]. To address the nonlinearity and non-stationarity of EEG signals, Tianjian Luo developed a common spatial pattern (CSP) selection method based on parallel genetic algorithms, enhancing classification accuracy and robustness [5]. In the domain of emotion recognition, Shashi Kumar G S et al. utilized wavelet transforms combined with KNN and SVM to classify emotional states in the DEAP dataset, showing promising results [6]. Jianhua Zhang et al. compared wavelet transforms and nonlinear dynamic analysis and found that a combination of kernel spectral regression and random forest performed best for emotion classification [7].

Research on EEG-based robot control has also made significant progress. Bousseta et al. designed a BCI system using an SVM classifier to control a robotic arm through motor imagery signals, achieving a control accuracy of 85.45% [8]. Yang An developed a hybrid BCI real-time control system that extracts EEG features using LSTM and CNN, combined with an actor-critic decision model, improving online control accuracy to 93.12% [9]. Additionally, Chutiwan Boonarchatong devised a hybrid algorithm incorporating EEG signals, accelerometers, and magnetometers to achieve continuous robot control [10]. Although substantial progress has been made in this field, challenges such as cross-domain adaptation, real-time processing, and spatiotemporal feature extraction remain to be addressed.

Existing methods still have certain limitations: most models rely on single features or static classifiers, making it difficult to adapt to the complexity of EEG signals; deep models often lack interpretability and have limited generalization capability; moreover, challenges remain in real-time processing and cross-domain adaptation. To address these issues, this study proposes a Rule-based Motor Imagery Interaction Model (Rule-MII) that integrates rule matching with multi-feature analysis to achieve interpretable and robust recognition of motor intentions and emotional states, outperforming previous methods in real-time performance and adaptability.

III. MOTIVATION.

In the development of future human-machine collaboration systems, enabling machines to understand users' intentions and emotional states—and to respond dynamically and adaptively—is critical to advancing both interaction intelligence

and user experience. Particularly in mixed reality (MR) environments, interaction between users and the virtual-physical space is no longer limited to explicit modalities such as physical manipulation or speech commands. Instead, systems are expected to "read" users through multimodal sensing channels, facilitating a more natural, continuous, and embodied interactive experience.

EEG signals, as a non-invasive physiological modality capable of reflecting a user's internal cognitive and emotional states, hold significant potential for human-computer interaction. However, most existing EEG-based interaction studies remain at the level of single-task control or static classification, with limited exploration into critical challenges such as multimodal signal fusion, intention-emotion coordination, and complex behavior generation. Furthermore, EEG applications in real-world scenarios suffer from practical limitations including low signal quality, poor cross-subject generalizability, and high real-time processing demands—factors that constrain its feasibility as a primary control signal in MR systems.

To address these challenges, this study aims to explore how EEG signals can be effectively modeled and applied in complex interactive settings, with a specific focus on the following key questions:

(1) How to improve EEG classification accuracy and robustness: By refining preprocessing and feature extraction, and introducing more advanced classification algorithms, the stability of recognition across different subjects and environmental conditions can be enhanced.

(2) How to achieve multimodal signal fusion: Building on motor imagery signals, emotional signals can be further integrated to enable more natural and intelligent robot control.

(3) How to optimize robot control performance: While ensuring classification accuracy and real-time performance, the model's adaptability and scalability across different application scenarios should be strengthened.

The core objective of this study is not only to improve the performance of EEG signal classification, but also to explore a new paradigm in which EEG serves as a fundamental interaction driver within immersive systems—thereby promoting the integration of cognitive intelligence and human-centered interaction in real-world applications.

IV. RULE-BASED MOTOR IMAGERY INTERACTION MODEL

To achieve the above goals, we propose the Rule-based Motor Imagery Interaction Model (Rule-MII Model). The model realizes accurate classification of motor imagery tasks and emotional states through rule matching and multidimensional feature extraction methods, and drives the robot to perform appropriate actions based on the classification results. The construction of the model is divided into several key steps: data pre-processing, rule base construction, rule matching and feature extraction, and robot control command generation and optimisation based on the classification results, as shown in Figure 1. By combining multimodal features, the model

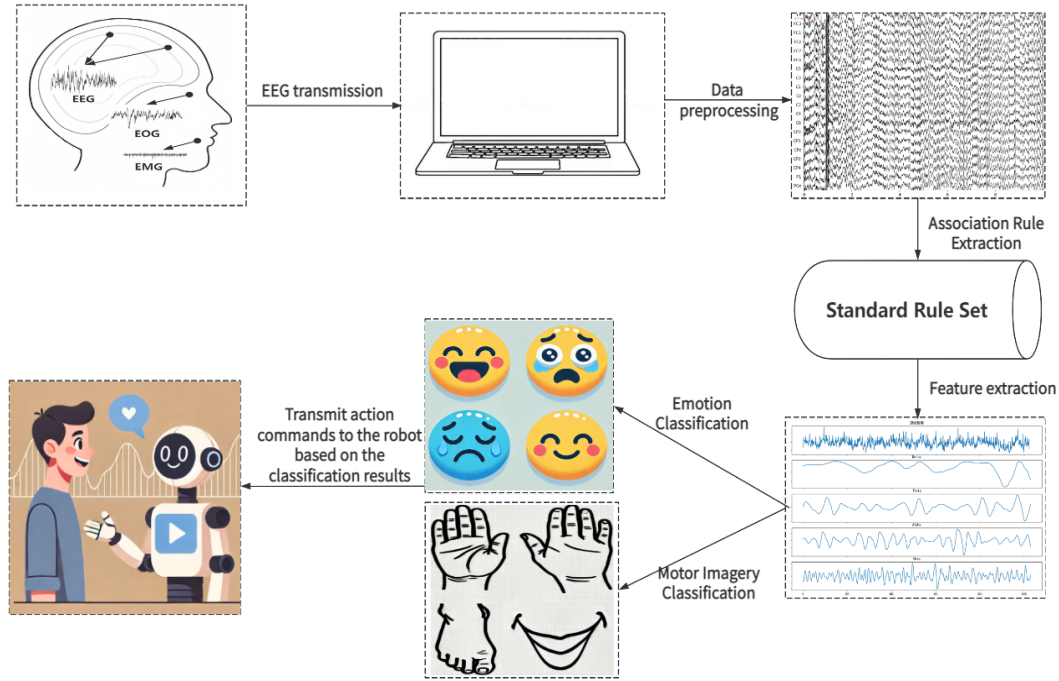


Fig. 1. EEG-based human-computer interaction.

improves the classification accuracy while ensuring the real-time responsiveness of the model.

A. Data Preprocessing

EEG signals typically contain low-frequency direct current components and high-frequency noise (such as electromyographic interference and power line noise), making band-pass filtering necessary. In this study, we applied a band-pass filter with a frequency range of 1-50 Hz. The low-frequency filtering was used to remove DC drift and low-frequency noise, while high-frequency filtering was applied to suppress electromyographic noise and power line interference. Since missing values may occur during EEG signal acquisition due to poor electrode contact or environmental interference, we employed a mode imputation method to handle these missing values. After completing the filtering and missing value processing, we applied Discrete Wavelet Transform (DWT) to perform multi-scale decomposition of the EEG signals and extract features across different frequency bands[16]. In this study, the Daubechies 4 (db4) wavelet was used as the mother wavelet to achieve multi-scale decomposition of the EEG signals, capturing characteristics within various frequency bands. The signals were decomposed into multiple levels of wavelet coefficients, including low-frequency approximation coefficients and high-frequency detail coefficients, corresponding to different frequency bands of the EEG signals, namely Delta (1-4 Hz), Theta (4-8 Hz), Alpha (8-13 Hz), Beta (13-30 Hz), and Gamma (30-50 Hz).

B. Standard Rule Set Construction for Quantized Signals

In the Rule-MII model, constructing the standard rule set is a key step for extracting EEG features and identifying signal patterns. After preprocessing and wavelet decomposition, peak and valley detection is performed on each frequency band of the EEG signal to capture significant changes in brain activity. The detection process uses a first-order difference-based extremum determination algorithm: a point is identified as a peak when $x_{i-1} < x_i > x_{i+1}$, and as a valley when $x_{i-1} > x_i < x_{i+1}$. To reduce noise interference, only valid extrema with an interval greater than a predefined threshold (typically 5% of the total sampling points) are retained. The detected extrema amplitudes are then normalized and quantized, mapping the amplitude range to the interval $[0, 1]$. Based on the relative amplitude percentages, the values are categorized into five discrete levels: Amplitude values representing more than 80% of the maximum magnitude value are classified as A, 80% – 60% as B, 60% – 40% as C, 40% – 20% as D, and 20% – 0% as E.

After quantization, a sliding window technique is applied to generate multi-scale combinational sequences. The window lengths are set to 2, 3, 4, and 5, with a 50% overlapping strategy to ensure the continuity of the sequences and preserve dynamic features. Windows of different lengths capture both short-term and long-term EEG variations, providing a rich data structure for subsequent frequent itemset mining.

After obtaining the combinatorial sequences generated by the sliding window, we use the FP-Growth algorithm for frequent itemset mining. Frequent itemsets are combinations

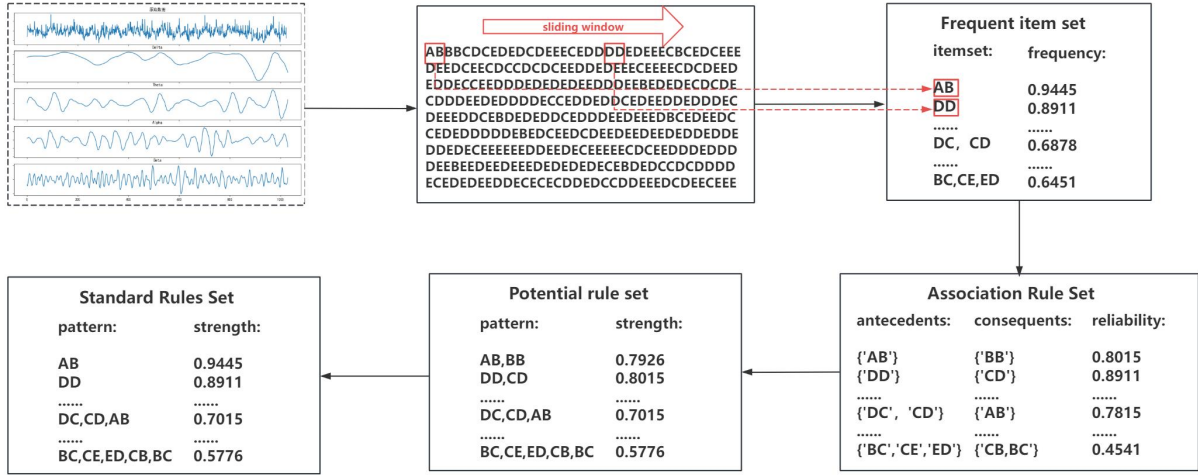


Fig. 2. Standard Rule Set Construction Process.

of symbols that appear frequently in a dataset, reflecting signal features that often co-occur in different time windows. In the FP-Growth algorithm, the frequent itemset mining process mainly consists of two steps: firstly, constructing the FP tree by scanning the dataset and counting the support degree of each symbol, and secondly, recursively extracting the conditional pattern base and the conditional FP tree from the FP tree, so as to discover all the possible frequent itemsets. After the frequent itemset mining is completed, the stage of association rule extraction is entered.

The purpose of association rule extraction is to uncover potential relationships between symbols from frequent itemsets that can reveal complex interaction patterns in EEG signals. Association rules are usually expressed in the form of 'If-Then', which describes the likelihood of the simultaneous occurrence of certain features under certain conditions. The extraction process of association rules is based on the frequent itemsets that have been generated in the previous steps. Firstly, the candidate rules (i.e., the set of association rules) are generated from the set of frequent items, and the support and confidence of each rule are calculated. The support is calculated based on the frequency of occurrence in the frequent item set, while the confidence level is determined by calculating the proportion of simultaneous occurrences of the antecedent and the consequent. Rules are retained only if both their support and confidence exceed a preset threshold, forming a potential rule set. In order to enhance the differentiation of rules, we removed the common rules between categories and retained the standard rule sets unique to each category, forming a standard rule base for different types of EEG activities. The construction of the standard rule base is shown in Figure 2.

C. Rule-Based Matching for Feature Extraction

After constructing the standard rule base, we matched new EEG signals with the rules in the base to extract features related to specific EEG activities. Its purpose is to transform the continuous EEG waveform into discrete feature patterns

by symbolizing and rule-structuring the time-series signals, thereby expressing complex brain activities in an interpretable and structured form. The feature matching process involves segmenting the preprocessed EEG signals into multiple sliding windows. Then, the combination sequences within each window are matched one by one with the rules in the base. Features such as the number of matching rules, the strength, density, repetition rate of the matched sequences, and polynomial relationships are extracted and compiled into a feature vector.

For each sliding window sequence S_i , the number of rules from the rule base that are satisfied is calculated. Let the rule base contain K rules, and the number of matching rules $N_{rules}(S_i)$ is given by:

$$N_{rules}(S_i) = \sum_{j=1}^K \text{Match}(S_i, R_j) \quad (1)$$

This feature represents the number of rules that are satisfied within the current sliding window. The matching strength of each rule in the sliding window sequence refers to the intensity or frequency of the features in the signal that align with the standard rule set within a specific time window. The strength of the matching sequence is defined as:

$$S(t) = \sum_{i=1}^N w_i \cdot R_i(t) \quad (2)$$

$S(t)$ represents the matching strength at time point t , where N is the number of rules in the current window, and w_i is the weight of rule i , indicating the importance of that rule. $R_i(t)$ is the matching result of rule i at time point t , typically 0 or 1, indicating whether the time point satisfies the rule. The density of the matching sequence represents the concentration of matched rules within the sliding window, and it is defined as the proportion of rule pairs that satisfy the matching condition within the sliding window:

TABLE I
ROBOT INTERACTION COMMANDS

Emotion	Motor Imagery	Robot Action	Emotion	Motor Imagery	Robot Action
Joy	Left hand	Smile and gently nod	Fear	Left hand	Quickly turn and raise palm
Joy	Right hand	Cheerfully clap	Fear	Right hand	Hug itself
Joy	Feet	Spin lightly and spread arms	Fear	Feet	Step back and issue a warning sound
Joy	Tongue	Cheer and raise arms	Fear	Tongue	Stop and look around
Sad	Left hand	Slowly extend hand and pause	Relaxation	Left hand	Slowly stretch arms and take a deep breath
Sad	Right hand	Gently pat its chest	Relaxation	Right hand	Gently clap and smile
Sad	Feet	Slowly step back and lower head	Relaxation	Feet	Lightly sway
Sad	Tongue	Sigh and sit down	Relaxation	Tongue	Deep breathing motion

$$D(S_i) = \frac{N_{\text{rules}}(S_i)}{L} \quad (3)$$

This density feature reflects the distribution of combination sequences within the sliding window, indicating the frequency of rule occurrences within a specific time window. The repetition rate is used to measure the frequency of repeated occurrences of symbol combinations within the sliding window. Let $\text{repeat}(R_j, S_j)$ represent the number of times the combination sequence repeats within the sliding window S_i , then the repetition rate is defined as:

$$R_{\text{repeat}}(S_i) = \frac{1}{K} \sum_{j=1}^K \text{repeat}(R_j, S_i) \quad (4)$$

The polynomial rate is used to quantify the variation of matching sequences across different window lengths, and its calculation can be expressed as:

$$P = \frac{\sum_{i=1}^n (N_i - N_1)}{N_{\text{total}}} \quad (5)$$

P represents the polynomial rate for the i -th window, N_i is the number of matching sequences in the i -th window, N_1 is the number of matching sequences in the first window, and N_{total} is the total number of matching sequences across all windows.

In addition to rule-matching features, both time-domain and frequency-domain features were integrated to comprehensively analyze motor imagery EEG signals. Time-domain features include statistical characteristics such as mean and standard deviation, which reflect the fluctuation patterns of the signal over time. For frequency-domain features, power spectral density (PSD) was used to quantify the energy distribution of the signal across different frequencies. Non-parametric estimation methods such as the Welch method were employed to effectively extract the PSD and capture the signal's frequency-domain characteristics. Finally, all extracted features were combined into a feature vector, which was then used for classification using a Support Vector Machine (SVM).

D. Robot Interaction Commands

The robot commands are designed to tightly integrate the classification results of the EEG signals with the robot action generation to achieve a real-time response based on the user's

EEG activities. First, the EEG signals are passed through an SVM classifier to output the corresponding emotional states and motor imagery results, such as the emotions of happiness, sadness, fear or relaxation, as well as the motor imagery of the left hand, right hand, foot or tongue. Each classification result is mapped to a specific robot action, forming a one-to-one mapping relationship.

The instruction library stores all predefined actions, and each classification label corresponds to a set of action instructions. During robot instruction generation, the output of the classification model is first parsed into emotional and motor imagery categories, and then the corresponding set of actions is selected from the instruction library. The generated instructions are then transmitted to the robot's action controller via a communication module, and the robot executes the appropriate actions based on the received instructions. The robot commands are shown in Table 1.

In the process of optimizing the robot control module, the response speed and accuracy of the system were first improved by optimizing and grading the action commands. The specific implementation involves decomposing complex actions into simple commands and setting priorities to ensure that critical actions are executed first. Suppose the classification result is C , and its output is a combination of emotional state and motor imagery, such as $C = (\text{Joy}, \text{Left_Hand})$. The action corresponding to each classification label is denoted as A_C . By defining a priority function $P(A_C)$, the priority function is formulated as:

$$P(A_C) = \begin{cases} \text{High,} & \text{if } C = (\text{Fear}, *) \\ \text{Medium,} & \text{if } C = (\text{Joy}, *) \text{ or } (\text{Sad}, *) \\ \text{Low,} & \text{if } C = (\text{Relaxation}, *) \end{cases} \quad (6)$$

This function ensures that high-priority actions are executed first. For a complex action A , such as "cheerful clapping," it can be decomposed into several simple actions, $A_1 = \text{raise hands}$ and $A_2 = \text{clap hands}$. In this way, the total execution time of the complex action, $T(A)$, can be expressed as the sum of the execution times of each sub-action: $T(A) = T(A_1) + T(A_2)$. By executing non-conflicting actions in parallel, the total execution time can be reduced.

V. EEG-CONTROL ROBOT INTERACTION EXPERIMENT RESULTS.

A. Dataset.

In this study, we used two widely applied public EEG datasets: BCI2a [13] and DEAP [14]. The BCI2a dataset contains EEG recordings from 9 subjects performing four types of motor imagery tasks: imagining movements of the left hand, right hand, both feet, and tongue. This dataset was collected using 22 EEG channels with a sampling rate of 250 Hz, and each task lasted approximately 4 seconds. It is primarily used for research on motor imagery classification. The DEAP dataset, used for emotion recognition tasks, includes EEG recordings from 32 subjects while they watched emotionally stimulating videos. The EEG signals were recorded through 32 channels with a sampling rate of 512 Hz. Emotional states were labeled along two dimensions: arousal and valence, and classified into four categories: HAPV (high arousal positive valence), LAPV (low arousal positive valence), HANV (high arousal negative valence), and LANV (low arousal negative valence), the emotion mapping is shown in Figure 3. Both datasets were randomly split into training and testing sets in an 8:2 ratio.

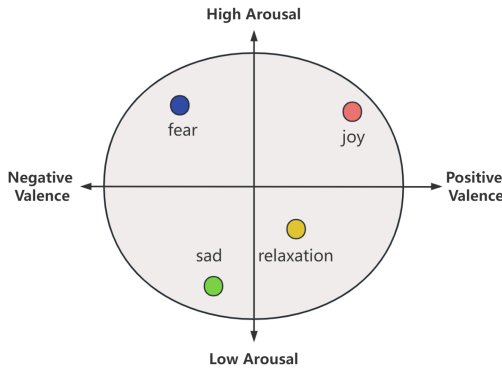


Fig. 3. Emotion Mapping in the Valence-Arousal Domain.

B. Motor Imagery Classification Results.

Motor Imagery Classification Results: In the BCI2a dataset, our experiment focused on classifying four types of motor imagery signals. We used SVM as the classifier and combined rule-matching features, time-domain, and frequency-domain features for feature extraction and classification. The experimental results are shown in Table 2.

The experimental results demonstrate that rule-matching features excel in capturing frequently occurring patterns in EEG signals. The SVM model utilizing only rule-matching features achieved an average classification accuracy of 89.3%. When combining rule-matching features with time-domain and frequency-domain features, classification performance improved significantly, the Rule-MII Model achieved an average classification accuracy of 91.0%. Among the four motor imagery tasks, the classification accuracy for left-hand movements was 92.1%, for right-hand movements 91.9%, for both

feet 90.2%, and for tongue movements 90.1%. This indicates that the combination of features effectively leverages the multidimensional information in EEG signals. In particular, rule-matching features help capture important patterns in complex signals, while time-domain and frequency-domain features provide additional detail on the dynamic changes within the signals.

In the comparative experiments with different feature sets, using only time-domain or only frequency-domain features resulted in relatively lower classification performance and failed to comprehensively capture the key characteristics of motor imagery signals. In addition, we compared the SVM model that integrates rule-matching, time-domain, and frequency-domain features with traditional time-frequency-based CSP-LDA and CNN models, as well as recent deep networks such as MSFF-SENet and EEG-TransNet. The CSP-LDA model achieved an average classification accuracy of 80.3%, while the CNN model reached 83.5%. MSFF-SENet and EEG-TransNet achieved improved accuracies of 85.6% and 85.3%, respectively, but still suffered from longer training times and a strong dependence on large-scale data. In contrast, the SVM model with fused rule-matching, time-domain, and frequency-domain features demonstrated superior performance in terms of accuracy, stability, and training efficiency.

C. Emotion Classification Results.

In the emotion classification task using the DEAP dataset, emotional states were divided into four categories: HAPV, LAPV, HANV, and LANV. The experimental results, as shown in Table 3, show that the Rule-MII model achieves an average classification accuracy of 92.7%, which is significantly better than the models using time-domain features or frequency-domain features alone, significantly outperforming models that used only time-domain or frequency-domain features. Specifically, the classification performance for HAPV and LANV was the best, with accuracies of 93.5% and 93.1%, respectively. This indicates that EEG signals in high arousal and low arousal states exhibit more distinct feature patterns, contributing to improved classification accuracy.

To further validate the effectiveness of the rule-matching features, we conducted comparative experiments, which showed that the SVM model combining rule-matching, time-domain, and frequency-domain features significantly outperformed the other models in the four-classification task. In contrast, the two-model approach based on saliency fusion proposed by Victor Delvigne et al. is less accurate. Although RNN and CNN have advantages in processing temporal data, RNN and CNN models do not perform as well as SVM models that combine multidimensional features in shorter time windows due to the complexity of sentiment signals and noise in EEG signals. The rule-matched feature SVM model, which combines time- and frequency-domain features, can fully exploit the multidimensional information in the emotion EEG signals, especially in capturing the key patterns in the emotion signals, which shows better robustness and accuracy,

TABLE II
MOTOR IMAGERY CLASSIFICATION RESULTS(%)

Model	Left Hand	Right Hand	Feet	Tongue	Average Accuracy
Time-domain Features + SVM	75.9	75.2	73.4	72.3	74.2
Frequency-domain Features + SVM	80.5	80.2	77.8	76.9	78.5
Rule-matching Features + SVM	90.7	89.9	88.5	88.2	89.3
Rule-MII Model	92.1	91.9	90.2	90.1	91.0
CSP-LDA [15]	81.0	79.8	78.9	80.5	80.3
ATCNet [16]	85.9	84.8	82.2	81.1	83.5
MSFF-SENet [17]	86.2	85.8	85.1	85.3	85.6
EEG-TransNet [18]	85.6	85.6	85.3	84.7	85.3

TABLE III
EMOTION CLASSIFICATION RESULTS(%)

Model	HAPV	LAPV	HANV	LANV	Average Accuracy
Time-domain Features + SVM	71.2	69.8	68.5	72.5	70.5
Frequency-domain Features + SVM	83.4	81.8	80.7	83.3	82.3
Rule-matching Features + SVM	92.5	92.0	90.1	92.7	91.82
Rule-MII Model	93.5	92.2	92.0	93.1	92.7
RNN+CNN [19]	79.2	76.8	77.3	80.0	78.47

proving the effectiveness and stability of multi-feature fusion in emotion classification.

D. Robot Control Experiment Results.

In this experiment, we integrated the results of motor imagery and emotion classification to apply them to robot control. The robot executed a series of compound actions corresponding to the user's emotional state and motor imagery signals. Motor imagery classification was performed using the BCI2a dataset, while emotional states were classified using the DEAP dataset, ultimately driving the robot to execute predefined actions based on the combined signals. The experimental results are shown in Table 4.

TABLE IV
ROBOT RESPONSE TIME(S)

Robot Commands	Left Hand	Right Hand	Feet	Tongue
Joy	2.1	2.0	2.3	2.4
Sad	2.2	2.1	2.4	2.5
Fear	2.3	2.2	2.5	2.6
Relaxation	2.0	2.1	2.2	2.3

We performed detailed measurements of the instruction generation time and response latency of the system to evaluate the real-time response capability of the robot after receiving EEG signals. The instruction generation time refers to the time from the output result of the classifier to the time when the robot generates the corresponding action instruction, and this period includes the processing of the motor imagery and emotion classification results as well as the instruction mapping process. The average classification result is 0.3 s. The action execution delay refers to the time taken by the robot from receiving the instruction to actually executing the action, which is on average 2 s due to the limitations of the computational power and memory capacity of the robot

processor in this experiment. Although there is some delay in the system's processing of motor imagery and emotional signals, the overall latency remains within 2.3 s, which meets the requirements of real-time control. The robot is able to execute the corresponding actions according to the classification results in a shorter time, showing a better interaction experience, and the delay has little effect on the user experience, which can achieve a smoother human-robot interaction.

VI. DISCUSSION.

The experimental results show that the Rule-MII model performs well on both motor imagery and emotion classification tasks, and that the robot is able to accurately execute the corresponding composite actions. This indicates that EEG signals can be used not only for single motor imagery or emotion classification, but also for more complex behavioral control through feature combination.

In the motor imagery task, the Rule-MII model achieves high classification accuracy, especially for left- and right-hand movements, with an accuracy of 92%. This suggests that EEG signal changes associated with motor imagery can be effectively captured through the extraction of rule-matching features. However, for more complex motor imagery tasks (such as tongue and feet movements), the classification accuracy decreases. This may be because such movements involve a broader range of neural activity and more complex signal patterns. In future work, more advanced feature extraction methods could be introduced to further improve classification performance. In the emotion classification task, the Rule-MII model also achieves satisfactory results, particularly for the HAPV and LANV emotional states. This indicates that, in these states, changes in the frequency-domain characteristics of EEG signals are more pronounced, making it easier for the model to capture emotional variation patterns. In contrast, the classification accuracy for LAPV and HANV is slightly

lower, reflecting the greater complexity of signal changes in these emotional conditions.

In the joint task of motor imagery and emotion classification, the experimental results show that the robot can execute complex compound actions based on the user's EEG signals. Compared with control schemes that rely solely on motor imagery or emotion classification, this joint control paradigm offers more flexible, context-aware, and individually adaptive behavior mapping. Specifically, motor imagery primarily provides intention information about "what action to perform," while emotional state provides modulation cues about "how the action should be performed." By integrating the two, the robot can not only accomplish command-level control, but also adapt action amplitude, speed, and interaction style according to the user's current emotional state, thereby achieving more natural and semantically appropriate behavioral responses. This demonstrates that combining motor imagery and emotional information for behavior mapping can effectively enhance the immersion of EEG-driven interaction systems and improve the quality of human-robot collaboration.

VII. CONCLUSION.

This study proposes a rule-based motor imagery interaction model (Rule-MII Model), which establishes a multimodal EEG recognition and behavior-mapping framework for immersive human-computer interaction. By combining rule matching with multi-feature extraction, the model can simultaneously recognize users' motor imagery and emotional states and map the results to embodied robot commands. Experimental results show that the proposed method achieves high accuracy and responsiveness in EEG signal processing, feature modeling, and compound behavior control, providing technical support for constructing context-aware and user-sensitive interaction systems in mixed reality environments. Compared with traditional BCI control schemes, the Rule-MII Model goes beyond simple action triggering by integrating motor intention and emotional information to enable multidimensional semantic understanding and behavior generation, promoting a shift in interaction paradigms from "command-driven" to "cognition-driven," and demonstrating the potential of EEG as an implicit input modality for supporting natural and personalized interaction in complex virtual-physical environments.

Future work will focus on improving the model's generalization capability and expanding its application scenarios. On the one hand, we plan to incorporate additional physiological signals—such as heart rate, galvanic skin response, and eye movements—to achieve more robust cross-modal fusion and continuous user state perception. On the other hand, we aim to validate the model in real-world augmented reality and intelligent environments, promoting the deployment of EEG-driven interaction systems in domains such as smart homes, rehabilitation, and immersive education.

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